By defn. of not belongs to, $x \notin A$ and $x \notin B$. By defn. of characteristic function, $f_A(x) = f_B(x) = 0$. $\therefore f_A(x) \cdot f_B(x) = 0$

$\boxed{f_{A \cup B}(x) = f_A(x) + f_B(x) - f_A(x) \cdot f_B(x)}$, $\because 0 + 0 - 0 = 0$ □

c) $f_{\bar{A}}(x) = 1 - f_A(x)$

Soln: Let, $x \in \bar{A}$. $\therefore f_{\bar{A}}(x) = 1$, by defn. of characteristic function

By Defn. 5 from Sec-2.2, $x \notin A$. $\therefore f_A(x) = 0$, by defn. of characteristic function. $1 = 1 - 0 \Rightarrow f_{\bar{A}}(x) = 1 - f_A(x)$.

Let, $x \notin \bar{A}$. $\therefore f_{\bar{A}}(x) = 0$, by defn. of characteristic function.

By Defn. of not belongs to, $\neg(x \in \bar{A})$ is True. By Defn. 5 from Sec-2.2, $\neg(x \notin A)$ is True. By Defn. of not belongs to, $\neg(\neg(x \in A))$ is True. By Double-negation Law of propositions, $x \in A$. By defn. of characteristic function, $f_A(x) = 1$.

$0 = 1 - 1 \Rightarrow \boxed{f_{\bar{A}}(x) = 1 - f_A(x)}$ □

d) $f_{A \oplus B}(x) = f_A(x) + f_B(x) - 2 f_A(x) f_B(x)$

Soln: By Prob 41 from Exercise 2.2, $A \oplus B = (A \cup B) - (A \cap B)$

$\Rightarrow A \oplus B = (A \cup B) \cap (\overline{A \cap B})$ (Prob 21(a) from Ex-2.2).

$\therefore f_{A \oplus B}(x) = f_{(A \cup B) \cap (\overline{A \cap B})}(x) = f_{(A \cup B)}(x) \cdot f_{(\overline{A \cap B})}(x)$ (Prob 73(a))

$= f_A(x) + f_B(x) - f_A(x) \cdot f_B(x)) \cdot f_{(\overline{A \cap B})}(x)$ (Prob 73(b))

$= (f_A(x) + f_B(x) - f_A(x) \cdot f_B(x)) \cdot (1 - f_{(A \cap B)}(x))$ (Prob 73(c))

$= (f_A(x) + f_B(x) - f_A(x) \cdot f_B(x)) \cdot (1 - f_A(x) \cdot f_B(x))$ (Prob 73(a))

$= f_A(x) - f_A^2(x) \cdot f_B(x) + f_B(x) - f_A(x) \cdot f$ (Wrong

Soln: Let, $x \in A \oplus B$. By Defn. of characteristic function, $f_{A \oplus B}(x) = 1$. By Defn. of symmetric difference from Ex-2.2, x belongs to exactly one of A or B. WLOG, Let, $x \in A$ but $x \notin B$. By Defn. of characteristic function, $f_A(x) = 1$ and $f_B(x) = 0$.